

AI for Engineers

Group Assignment: Middle America Job and Application Agent

Complete Assignment Packet | Group Size: 5 Students

1. Assignment Overview

Design and implement an AI agent that autonomously browses the web to find AI Engineer jobs at mid-sized "Middle America" companies (not big tech or startups), then simulates applicant workflows by tailoring resumes and cover letters. The agent must filter and rank jobs, then generate personalized applications.

Group size: 5 students

Deliverables: Code repository, report (8-10 pages), recorded demo (8-10 min), peer assessment, individual reflections.

Learning Objectives

Build production-ready agent skills across:

- Web interaction and scraping
 - LLM personalization and prompt engineering
 - Constraint-based filtering and ranking
 - Hiring-focused evaluation and benchmarking
 - Ethical AI and bias analysis
-

2. Agent Pipeline & Scenario

Pipeline Steps

1. Search job boards; extract title, company, location, skills, salary, and URL.
2. Filter: Blacklist FAANG+; exclude startups (<50 employees heuristic).
3. Rank top 10 by skill match, location preference, and recency.
4. Applicant Simulation: Tailor provided sample resume + cover letter for top 3 jobs.

Sample Resume: Generic AI Engineer profile (3-5 years experience, Python / TensorFlow / MLflow skill set).

Group Roles

5. **Product Lead:** Requirements, constraints, sample resume definition.
6. **Agent Architect:** Pipeline flow and architecture design.
7. **Web Engineer:** Scraping and parsing using APIs such as SerpAPI.
8. **LLM Engineer:** Ranking logic and tailoring prompt engineering.
9. **Eval Lead:** Benchmark experiments and bias analysis.

3. Required Components

3.1 Design Document (2-3 pages)

- Pipeline diagram (visual flow of all stages).
- Filter heuristics with justification.
- Tailoring workflow description.

3.2 Implementation

- Complete end-to-end pipeline: Search → Filter → Rank → Tailor.
- Logging of all decisions and rationales at each stage.

3.3 Evaluation (Core)

Hiring Simulation (Priority):

- Construct a ground-truth 20-job benchmark (10 interview-worthy, 10 rejects).
- Have 3 humans score the agent shortlist: "Interview? (Yes/No)".
- Compute Precision@10 and interview yield.

Additional Evaluation:

- Tailoring: Human score (1-5) compared to manual baseline.
- Adaptation: Filter toggle test (e.g., "Iowa-only" mode).

3.4 Report (8-10 pages)

10. Introduction and requirements.
11. Design and implementation details.
12. Evaluation results (hiring simulation focus).
13. Bias and ethics analysis (tailoring fairness).
14. Appendix: Agent trace.

3.5 Recorded Demo (8-10 minutes)

- Fresh query run with screen share + voiceover.
- Filter toggle demonstration (e.g., Texas → adapt).
- Top job explanation: "85% match, Texas, <30 days."
- Tailor sample resume for top 3 jobs; narrate changes.
- One failure case + how the team fixed it.
- Unlisted video URL with timestamped segments.

3.6 Individual Reflections (1 page each)

- Personal role and contributions to the project.

- Reflection on hiring equity and the ethical impact of the agent.

4. Student Submission Checklist

Before submitting, each group must verify all items below are included and complete. Use the self-check column to confirm.

Deliverable	Requirements / What to Include	Self-Check (✓)
Design Doc (2-3 pages)	Pipeline diagram; filter heuristics explained; tailoring workflow description	
Implementation (Code)	End-to-end pipeline: Search → Filter → Rank → Tailor. Include logging of decisions and rationales.	
Evaluation Results	20-job benchmark (10 interview-worthy / 10 rejects); Precision@10 + interview yield metrics; tailoring human scores (1-5); filter toggle experiment.	
Report (8-10 pages)	Sections: Intro, Design/Impl, Eval results (hiring sim focus), Bias/Ethics, Appendix (agent trace).	
Recorded Demo (8-10 min)	Fresh query run; filter toggle demo; top-job explanation (match %, location, recency); tailoring narration for top 3 jobs; one failure + fix. Unlisted video with timestamps.	
Individual Reflections	1 page per student: personal role/contributions; reflection on hiring equity impact.	
Peer Assessment	Each member submits scores and comments for all teammates on contribution, communication, and quality.	
Code Repository	Shared GitHub/GitLab repo with README, requirements.txt / environment.yml, and clear run instructions.	

Tip for students: TAs will run your code from scratch using only your README. If it fails to run, implementation points will be deducted.

5. Role-Specific Accountability

Each student is primarily responsible for their assigned role deliverables. TAs will verify individual contributions through the peer assessment, reflections, and git commit history.

Role	Primary Deliverables	TA Verification Points
Product Lead	Requirements doc, sample resume, constraint definitions.	Is resume realistic (3-5 yr AI Engineer)? Are constraints (FAANG blacklist, ≥50 employees) clearly defined and enforced?
Agent Architect	Pipeline design, architecture diagram, integration plan.	Does architecture diagram match actual code structure? Is separation of concerns maintained?
Web Engineer	Scraping/parsing module, API integration (SerpAPI), data extraction.	Does job extraction reliably capture title, company, location, skills, salary, URL? Any rate-limit handling?
LLM Engineer	Ranking logic, tailoring prompts, personalization output.	Are ranking criteria (skill/location/recency) applied correctly? Are tailored docs meaningfully differentiated?
Eval Lead	Benchmark dataset, metrics computation, bias analysis.	Is benchmark sound (20 jobs, labeled)? Are Precision@10 and yield computed correctly? Is bias section substantive?

Note: The overall group grade is shared, but peer assessment scores can adjust individual grades by up to ± 5 points.

Please upload your code in Github with a proper readme file on how to install packages and run your code.